# Storm Damage Risk Analytics: Visual Analytics of Recurring and Catastrophic Severe-Weather Damage across Texas Counties

Project proposal (v2) · CSCE 679 Data Visualization · Team 8
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## 1. Research Background and Motivation

### Domain

Our domain is **property insurance in Texas** and the damage caused by severe convective storms: hail, thunderstorm wind and tornadoes. These storms matter to insurers: between 2019 and 2025, Texas insurers paid **$56.6 billion** on homeowners policies, of which wind and hail made up **61%** [1], and the median county average premium rose **70%** [2].

### Problem statement

**Insurance underwriters and risk analysts need to understand why historical severe-weather losses differ across geographic regions.** Total reported damage and storm frequency can be misleading, because they do not distinguish persistent weather exposure from isolated catastrophic events. Our preliminary analysis of NOAA's Storm Events Database [3] (25,867 Texas hail, wind and tornado reports, 2015–2025) shows how strong this effect is. **Tarrant and Bexar counties have almost the same NOAA-reported property damage ($1.44B vs $1.37B)**, but Tarrant's came from three storms of more than $350M in 2016 and 2021 plus damaging storms in every year, whereas **99% of Bexar's came from one hailstorm** on 12 April 2016 (Figure 1, left). Of the 57 counties with at least $5M of reported damage, **48 received at least half of it on a single storm day**. Frequency misleads as well: tornadoes were 6% of reports but 23% of damage, and thunderstorm wind 40% of reports but 4% of damage (Figure 1, right). Finally, the record is incomplete: 15% of reports leave damage blank and 64% report $0, and NOAA damage is an estimate, not an insured loss [4].

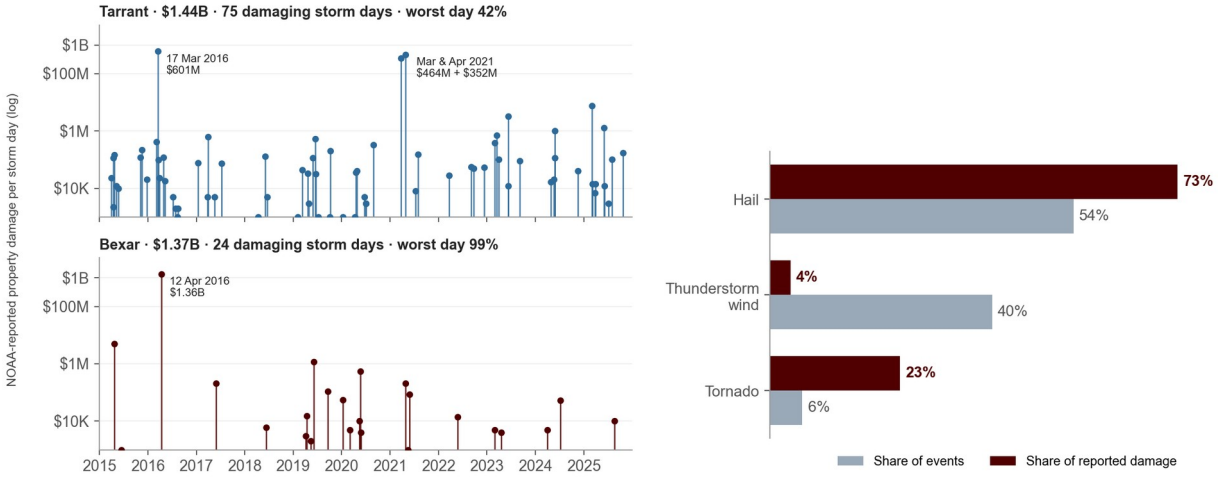



*Figure 1. Left: NOAA-reported property damage per storm day (log scale), 2015–2025, for two counties with similar totals. Right: each hazard's share of Texas storm reports and of reported damage, 2015–2025. Team analysis of [3].*

### Goal

Insurers already use commercial tools to estimate repairs, rebuilding costs and catastrophe losses, but none of them shows the observed storm record behind a county's losses (Section 2), and catastrophe-model vendors note that this record is short and under-reports storms in rural areas [5]. **Our goal is an interactive visualization that helps analysts compare counties, identify the hazards responsible for historical damage, and distinguish recurring storm impacts from rare extreme events.** Seeing frequency, severity, hazard, time and individual events together suits coordinated views better than separate tables or a single risk score.

## 2. Design Requirements

### Target users

Our primary users are **property-insurance analysts**: **underwriting analysts**, who examine the storm history behind a location; **catastrophe-risk analysts**, who ask whether a county's damage is recurring or driven by a few extreme events; and **portfolio-risk analysts**, who compare counties in a book of business. Our secondary users are **homeowners and communities**, for example a homeowner deciding whether recurring local hail justifies an impact-resistant roof, for which TDI encourages insurers to offer actuarially supported discounts [6].

### Challenges with current insurance tools

Insurers rely on three widely used commercial tools (Table 1). Based on their public documentation, each answers a different question, and each leaves a gap for an analyst who must explain regional storm losses. We will check these challenges in the interviews.

| Tool | What it answers | Challenge for regional storm analysis | Leads to |
|---|---|---|---|
| **Verisk Xactimate [7]** | Cost to repair one damaged structure, from line-item, locally priced estimates | Works claim by claim after a loss; does not show how often, or by which hazard, a county is struck, so counties cannot be compared | R1, R3, R5 |
| **Verisk 360Value [8]** | Cost to rebuild a home (insurance-to-value) at quote and renewal | Values the building, not the hazard; storm history must come from elsewhere; raw damage totals ignore how built-up a county is | R1, R6 |
| **Moody's RMS Risk Modeler [9]** | Modeled portfolio losses from simulated events: average annual loss, exceedance curves | Averages hide whether losses recur or come from the tail; results are hard to trace to real storms and years; licensed specialist tool, and vendor models can differ [10] | R2, R4, R5, R6 |

*Table 1. Commercial tools and the requirements their gaps motivate. Our system complements them; it does not price repairs or model risk.*

### Interacting with target users

Following feedback on our presentation, we will ground the requirements in **semi-structured interviews** before building the prototype. We plan seven: two practitioners (an underwriter or claims adjuster), one catastrophe or actuarial analyst, one public-sector reviewer (TDI or the Office of Public Insurance Counsel), two homeowners or community members, and one pilot with risk-management and insurance faculty or students at Texas A&M; each member conducts at least one. Each 45-minute session covers (1) **workflow** and the tools used today, (2) **pain points**, including what those tools cannot show, (3) a **think-aloud** on an early prototype with one task per requirement, and (4) **ranking** the requirements. Example questions: "How do you tell recurring storm damage from a one-off catastrophe?" and "Do storm records match what you see in claims?" We will code notes into an affinity diagram and revise R1–R6. To prepare the guide, we asked an AI assistant to role-play users of the three tools; these simulated answers only shaped our questions and are **not reported as findings**. Interviews are being scheduled and **no results are reported here**.

### Design requirements

- **R1 · Geographic loss comparison.** Select two or more counties and compare storm frequency, total reported damage and average damage per event, with the same period and hazard filters applied to all. *Why does County A have higher historical storm damage than County B?*
- **R2 · Recurring vs. catastrophic loss.** Compare mean and median damage per event, identify the largest events, show the share of damage from the top 1 or 5 events and the distribution across events. *Does this county experience damaging storms regularly, or is most damage caused by a few extreme events?*
- **R3 · Hazard contribution.** Break events down by hail, tornado and thunderstorm wind, compare each hazard's share of frequency and damage, and contrast the most frequent with the most damaging hazard. *Is damage mainly associated with hail, tornadoes or wind?*
- **R4 · Temporal loss pattern.** Compare 5-, 10- and 20-year periods, examine annual frequency and damage, and flag unusually high-damage years. *Has this county had repeated damaging storms over the last decade, or were losses concentrated in one or two years?*

- **R5 · Storm event investigation.** For a county, list its most damaging events with date, location, hazard, magnitude and damage, linked to county totals and trends. *Which specific storms explain the unusually high damage?*
- **R6 · Data context and interpretation.** Distinguish NOAA's estimated damage from insured claims, flag missing or zero damage, give context for differently developed counties, and never present history as a risk score or premium advice. *Can I trust this comparison, and what limits apply before an insurance decision?*

### Data

All data are public (Table 2). Storm events are assigned to counties by FIPS code and aggregated by county, year, hazard and storm day (all reports in a county on one date), because NOAA often splits one storm into many reports. Blank and zero damage are kept distinct, and dollar values will be shown in nominal and CPI-adjusted terms.

| Dataset | Unit · years | Key fields | Role |
|---|---|---|---|
| **NOAA NCEI Storm Events [3]** | Event → county · 2006–2025 | Hail, thunderstorm wind, tornado; date, location, magnitude, damage estimate, narrative | Storm history (R1–R5) |
| **TDI, homeowners losses by county [1]** | County × year · 2019–2025 | Paid losses by type (wind & hail, water, fire, other); policies in force | Insured-loss reference (R6) |
| **Census ACS 5-year [11]** | County | Housing units, owner-occupied homes, median home value | Development context (R1, R6) |
| **TDI premiums and HB 2067 reasons [2, 12]** | County / ZIP · 2019–2026 | Average premium; nonrenewal and declination reasons (e.g., roof, wind/hail) | Market context |

*Table 2. Datasets. TDI losses cover homeowners policies only (not TWIA wind/hail); NOAA damage is an estimate, not insured loss [4].*

### Data visualization tasks

Using Brehmer and Munzner's typology [13], each requirement maps to an abstract task and a planned encoding: **R1 → compare** counties side by side (juxtaposed bars under shared filters, linked to a map); **R2 → summarize a distribution and find outliers** (ranked-event curve of cumulative damage, with mean, median and top-1/top-5 shares); **R3 → compare proportions** (paired bars of each hazard's share of events and of damage); **R4 → identify trends and spikes** (annual bars with 5-, 10- and 20-year windows and flagged high years); **R5 → locate and look up** events (sortable table with map points and narratives); **R6 → annotate** reliability (missing-data badges, TDI paid losses as a reference line, per-housing-unit views). Selecting a county, hazard or period in one view filters all others.

## 3. Related Work

### Existing systems and research

**Property-risk estimation from hazard loss data.** Mostafiz et al. estimate census-block property-loss risk from hail, tornado, lightning and extreme cold in Louisiana using losses derived from NOAA Storm Events, and note that most of those losses are unverified estimates [14]. FEMA's National Risk Index computes expected annual loss for 18 hazards in every county and tract and summarises it as composite scores [15]. Gall et al. document six biases in hazard-loss databases, including NOAA Storm Events [4]. These works estimate risk statistically but do not support interactive exploration, and averages such as expected annual loss hide whether losses recur.

**Visual analytics for weather and hazard risk.** Li et al. present Havior, a visual analytics system for heat-risk management that combines model output with news reports processed by a large language model, evaluated with six domain experts [16]. Navarra et al. co-develop with practitioners a pipeline that detects extreme-weather impacts from social media for impact-based warnings [17]. Both show the value of expert-centred design, but neither addresses insurance or historical storm records.

**Public and commercial tools.** NOAA offers a Storm Events search [3], and TDI publishes maps of losses and premiums by county and, since September 2026, nonrenewal reasons by ZIP code [1, 2, 18]; each shows one

measure at a time. Commercial tools price repairs, value homes or model losses (Table 1) but do not show the observed record.

### Potential novelty

We found no tool that shows whether a county's observed storm damage is recurring or driven by a few extreme events. Our contribution is modest: **one coordinated workflow for insurance analysts that compares counties under the same filters, separates recurring from catastrophic damage, attributes it to hazards and individual storms, and keeps reported damage, insured loss and data gaps visibly distinct.** We do not propose a catastrophe or pricing model; patterns are presented as associations, not causes.

## 4. Development Plans

**Preliminary work.** An initial interactive mock-up on integrated NOAA, TDI and Census data (D3.js and Three.js) is online at https://insurviz.web.app, with code at github.com/hendrosetyawan/insurviz. It is a probe for the interview think-alouds and will be revised toward R1–R6.

### Task assignments

| Member | Role | Responsibilities |
|---|---|---|
| **MD Imtiaz Mahi** | Data integration & context (R6) | NOAA 2006–2025, TDI and ACS joins; storm-day aggregation; data-quality flags and caveats |
| **Aadi Mahajan** | County comparison (R1, R3) | Side-by-side county views with shared filters; hazard share views |
| **Hendro Setyawan** | Recurring vs. catastrophic (R2, R4) | Ranked-event and distribution views; 5/10/20-year timelines |
| **Abhinav Sheshadri** | Event investigation (R5) | Event table, map and narratives; coordinated selection and integration |

*Table 3. Roles. All four members conduct interviews and share design decisions, evaluation, report writing and presentations.*

### Timeline

| When | Milestone |
|---|---|
| **October** | User interviews (n = 7); affinity analysis; finalise R1–R6 |
| **Oct–Nov** | Integrate NOAA 2006–2025, TDI and Census data; storm-day and county × year tables |
| **November** | Prototype of R1–R3, then R4–R6; system integration |
| **Nov–Dec** | Think-aloud sessions with interviewees; refine design |
| **December** | Final system, evaluation summary, report and presentation |

*Table 4. Timeline; dates will be aligned with the course checkpoints.*

**Use of AI.** We used an AI assistant to brainstorm domains, refine the problem statement, find data sources, plan literature searches and role-play users of commercial tools to draft interview questions. Simulated answers were not used as evidence. All facts and figures were checked against NOAA, TDI, Census, vendor documentation and the original papers, and the team made all design decisions. Each member who used AI will upload their logs and reflection separately, as required.

## References


**[1]** Texas Department of Insurance. Texas homeowners losses by county (2019–2025). https://www.tdi.texas.gov/consumer/homeowners-losses-by-county.html (data updated 22 June 2026).

**[2]** Texas Department of Insurance. Texas homeowners insurance market overview: average premium by county, 2019–2025. https://www.tdi.texas.gov/general/texas-homeowners-insurance-market-overview.html

**[3]** NOAA National Centers for Environmental Information. Storm Events Database, details files. https://www.ncei.noaa.gov/stormevents/

**[4]** M. Gall, K. A. Borden, S. L. Cutter. When do losses count? Six fallacies of natural hazards loss data. Bulletin of the American Meteorological Society 90(6):799–810, 2009. doi:10.1175/2008BAMS2721.1

**[5]** T. Sabbatelli. If a tree falls in the forest: overcoming challenges in U.S. severe convective storm observations. Moody's, 25 November 2025. https://www.moodys.com/web/en/us/insights/insurance/overcoming-challenges-in-us-severe-convective-storm-observations.html

**[6]** Texas Department of Insurance. Commissioner's Bulletin B-0008-26 (2026), on premium discounts for loss mitigation including impact-resistant roofing. https://www.tdi.texas.gov/bulletins/2026/commissioners-bulletin-b-0008-26.html

**[7]** Verisk (Xactware). Xactimate: property claims estimating. https://www.xactware.com/home/products/xactimate.html, accessed October 2026.

**[8]** Verisk. Verisk Insurance Solutions expands 360Value property prefill to help streamline residential replacement cost estimation. https://verisk.com/company/newsroom/verisk-insurance-solutions-expands-360value-property-prefill-to-help-streamline-residential-replacement-cost-estimation, accessed October 2026.

**[9]** Moody's. Introducing Moody's RMS North America Severe Convective Storm HD Models. December 2025. https://www.moodys.com/web/en/us/insights/insurance/introducing-moodys-rms-north-america-severe-convective-storm-hd-models.html

**[10]** Guy Carpenter. Managing catastrophe model uncertainty: issues and challenges. December 2011. https://guycarp.com/insights/2011/12/managing-catastrophe-model-uncertainty--issues-and-challenges--p0.html

**[11]** U.S. Census Bureau. American Community Survey 5-year estimates, 2020–2024, tables B25001, B25003, B25077.

**[12]** Texas Department of Insurance. 89R HB 2067 datasets: policy counts; actual declinations, cancellations and nonrenewals; reasons. data.texas.gov (vyxq-akit, m7yx-zxf2, 3efz-d6qn), accessed October 2026.

**[13]** M. Brehmer, T. Munzner. A multi-level typology of abstract visualization tasks. IEEE Transactions on Visualization and Computer Graphics 19(12):2376–2385, 2013. doi:10.1109/TVCG.2013.124

**[14]** R. B. Mostafiz, C. J. Friedland, R. V. Rohli, M. Gall, N. Bushra, J. M. Gilliland. Census-block-level property risk estimation due to extreme cold temperature, hail, lightning, and tornadoes in Louisiana, United States. Frontiers in Earth Science 8, 2020. doi:10.3389/feart.2020.601624

**[15]** C. Zuzak, M. Mowrer, E. Goodenough, J. Burns, N. Ranalli, J. Rozelle. The National Risk Index: establishing a nationwide baseline for natural hazard risk in the US. Natural Hazards 114(2):2331–2355, 2022. doi:10.1007/s11069-022-05474-w

**[16]** H. Li, W. Kam-Kwai, Y. Luo, J. Chen, C. Liu, Y. Zhang, A. K. H. Lau, H. Qu, D. Liu. Save it for the "hot" day: an LLM-empowered visual analytics system for heat risk management. IEEE Transactions on Visualization and Computer Graphics 31(10):8928–8943, 2025. doi:10.1109/TVCG.2025.3586689

**[17]** C. Navarra, K. Kucher, T.-S. Neset, C. G. Villaro, F. Schück, J. Unger, K. Vrotsou. Leveraging visual analytics of volunteered geographic information to support impact-based weather warning systems. International Journal of Disaster Risk Reduction 126:105562, 2025. doi:10.1016/j.ijdrr.2025.105562

**[18]** Texas Department of Insurance. TDI launches new transparency tool showing why some home policies are declined, canceled, or not renewed. News release, 23 September 2026. https://www.tdi.texas.gov/news/2026/tdi09232026.html